

AI-POWERED LEARNING SKILLS PLATFORM

ABSTRACT

The AI-Powered Learning Skills Platform is a student-centered educational support system designed to convert handwritten or image-based notes into accessible and interactive study material. Many students face difficulty in revising handwritten notes because of poor legibility, lengthy content, and lack of accessibility support. This project addresses those challenges by combining OCR (Optical Character Recognition), AI-based summarization, text-to-speech, and study-material generation into a single prototype application.

The platform is built using Python and Streamlit. It allows users to upload an image of notes, extract text using Tesseract OCR, and generate a concise summary. The generated summary can also be read aloud using gTTS, making the system useful for auditory learners and for students who prefer audio-based revision. In addition to summarization, the system generates quiz questions, flashcards, study tips, and important keywords, thereby converting passive notes into active learning material.

The platform also provides downloadable outputs for summary, quiz, and flashcards, allowing students to save study content for later use. This project demonstrates how AI-powered educational tools can improve accessibility, revision efficiency, and student engagement through a simple web-based interface.

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CHAPTER 1 – INTRODUCTION

Students often rely on handwritten notes or captured note images for study and revision. However, revising from handwritten notes can be difficult due to poor handwriting, cluttered formatting, and the absence of accessibility support such as audio reading. In addition, students usually spend significant time converting long notes into summaries, revision points, and self-assessment material.

The AI-Powered Learning Skills Platform was developed to solve this problem by transforming uploaded note images into useful learning resources. The system combines OCR, AI summarization, audio reading, and study-content generation in one application. Instead of only extracting text from an image, the platform supports active learning by generating summaries, quizzes, flashcards, study tips, and keywords.

This project is also useful from an accessibility perspective. Students who prefer listening to study material or who find long written notes difficult to revise can benefit from the summary audio and concise study resources generated by the system.

CHAPTER 2 – PROBLEM STATEMENT

Students face several difficulties while studying from handwritten or image-based notes:

- Handwritten notes may be difficult to read because of unclear handwriting or formatting.
- Extracted OCR text can contain noise or formatting issues.
- Long notes are time-consuming to revise manually.
- Students often need additional study support such as summaries, quizzes, and flashcards.
- Traditional note images do not provide audio learning support.
- There is a need for a single platform that can convert raw note images into accessible, interactive study material.

The problem addressed by this project is how to create a simple AI-based learning platform that helps students convert uploaded notes into readable, summarized, audible, and revision-friendly study resources.

CHAPTER 3 – OBJECTIVES

The main objectives of the project are:

1. To develop a platform that allows students to upload note images.
2. To extract text from uploaded notes using OCR.
3. To generate a concise summary from the extracted text.
4. To convert the summary into speech for audio-based learning.
5. To generate quiz questions from the summarized content.
6. To generate flashcards for active recall and quick revision.
7. To generate study tips and important keywords from the content.
8. To allow downloading of generated study material.
9. To build a simple and accessible Streamlit-based prototype for student learning support.

CHAPTER 4 – EXISTING SYSTEM / LITERATURE REVIEW

Several existing tools provide OCR, note management, or AI assistance, but they do not fully support the complete learning workflow needed by students.

Existing Approaches

- OCR tools can extract text from images, but they usually stop at text extraction.
- Note-taking platforms help organize notes but may not automatically generate study material.
- AI summarization tools can summarize text, but they may not be integrated with OCR and text-to-speech in a single learning system.

Limitations of Existing Systems

- Lack of a complete OCR + summary + speech + quiz + flashcard workflow
- Limited support for transforming personal handwritten notes into study material
- Lack of integrated accessibility features for audio-based revision
- Students still need to manually create questions, flashcards, and revision points

Need for the Proposed System

There is a need for a unified platform that can:

- accept uploaded notes
- extract text automatically
- generate useful learning material
- support audio learning
- save time during revision

CHAPTER 5 – PROPOSED SYSTEM

The proposed system is an AI-powered study support platform that transforms note images into interactive study content.

Working of the Proposed System

1. The user uploads an image of notes.
2. The image is processed using Tesseract OCR to extract text.
3. The extracted text is displayed in the app.
4. The extracted text is summarized using Ollama.
5. The generated summary is converted into speech using gTTS.
6. Based on the summary, the system generates:

- quiz questions
- flashcards
- study tips
- keywords

7. The generated outputs can be downloaded for later use.

Benefits of the Proposed System

- Reduces revision time
- Makes handwritten notes easier to use
- Supports audio learning
- Encourages active recall using quizzes and flashcards
- Helps students revise more effectively from their own notes

CHAPTER 6 – SYSTEM DESIGN AND ARCHITECTURE

System Architecture

- Input Layer
 - User uploads handwritten or printed note image
- Processing Layer
 - Image is read using Pillow
 - OCR is performed using PyTesseract / Tesseract OCR
 - Extracted text is summarized using Ollama
- Learning Support Layer
 - gTTS converts summary into audio
 - Quiz questions are generated
 - Flashcards are generated
 - Study tips are generated
 - Keywords are extracted
- Output Layer
 - OCR text displayed
 - Summary displayed
 - Audio player available
 - Quiz / flashcards / study tips / keywords displayed
 - Download buttons for summary, quiz, and flashcards

Data Flow

Image Upload → OCR Extraction → Extracted Text → Summary → Audio + Quiz + Flashcards + Study Tips + Keywords → Downloadable Outputs

CHAPTER 7 – TECHNOLOGIES USED

7.1 Python

Python is the core programming language used to build the application logic and integrate the OCR, AI, and text-to-speech components.

7.2 Streamlit

Streamlit is used to build the user interface of the application. It allows quick development of interactive web apps with buttons, file upload, text display, and audio playback.

7.3 PyTesseract and Tesseract OCR

PyTesseract acts as the Python wrapper for Tesseract OCR. Tesseract OCR is used to extract text from uploaded note images.

7.4 Pillow

Pillow is used to open and process the uploaded image files before OCR is performed.

7.5 Ollama

Ollama is used to generate summaries and learning content such as quiz questions, flashcards, study tips, and keywords based on the extracted text.

7.6 gTTS

gTTS (Google Text-to-Speech) is used to convert the generated summary into audio, allowing the user to listen to the summarized notes.

CHAPTER 8 – IMPLEMENTATION AND MODULES

The application was implemented in a single Streamlit app file, `app.py`, supported by required libraries.

8.1 Upload Notes Module

This module allows the user to upload an image of notes in supported image formats.

8.2 OCR Text Extraction Module

The uploaded image is processed using PyTesseract to extract the note content as text.

8.3 Summary Generation Module

The extracted text is passed to Ollama, which generates a concise summary of the notes.

8.4 Audio Reader Module

The generated summary is converted into speech using gTTS, and the resulting audio is played inside the application.

8.5 Quiz Generation Module

This module generates quiz questions from the summarized content to support self-assessment.

8.6 Flashcard Generation Module

This module generates flashcards from the summarized content for active recall and revision.

8.7 Study Tips Module

This module provides short study tips or revision suggestions based on the content.

8.8 Keywords Module

This module extracts important keywords or key points from the summarized content.

8.9 Download Module

The application includes download buttons to save:

- summary
- quiz
- flashcards

CHAPTER 9 – RESULTS AND OUTPUTS

The developed application was tested locally with sample note images. The system successfully performed the following tasks:

1. Accepted note image upload through the Streamlit interface.
2. Extracted text from the uploaded image using OCR.
3. Displayed the extracted text in the application.
4. Generated a summary from the extracted content.
5. Converted the summary into audio and played it in the app.
6. Generated quiz questions from the summary.
7. Generated flashcards for revision.
8. Generated study tips.
9. Extracted important keywords.
10. Allowed the user to download the generated study material.

Screenshots to include in this chapter

- Home / upload screen
- OCR output screen
- Summary output screen
- Audio reader screen

- Quiz output screen
- Flashcards output screen
- Study tips output screen
- Keywords output screen
- Download buttons screen

CHAPTER 10 – ADVANTAGES, LIMITATIONS, AND FUTURE SCOPE

Advantages

1. Converts handwritten or image-based notes into readable text.
2. Reduces study time by generating summaries automatically.
3. Supports audio learning through text-to-speech.
4. Helps students practice active recall using quizzes and flashcards.
5. Provides additional study support using keywords and study tips.
6. Combines multiple learning-support features in one platform.
7. Simple and easy-to-use interface built with Streamlit.

Limitations

1. OCR accuracy depends on the quality of the uploaded image and handwriting clarity.
2. The current version works as a local prototype and does not include user login or cloud storage.
3. The quality of generated content depends on the quality of extracted text.
4. The system currently uses gTTS for speech output, which is basic compared to advanced voice tools.
5. Advanced reminder and analytics features are not included in the implemented prototype.

Future Scope

1. Use more advanced natural-sounding speech technology such as ElevenLabs for improved voice quality.
2. Add study reminders and learning analytics features.
3. Improve OCR cleanup for difficult handwriting.
4. Add a more advanced AI study companion / question-answer feature.
5. Extend the system with cloud storage and user accounts.
6. Add support for more file types such as PDF notes.

CONCLUSION

The AI-Powered Learning Skills Platform successfully demonstrates how OCR, AI summarization, text-to-speech, and study-content generation can be combined into a single educational support system. The platform converts note images into accessible digital learning material and improves the revision process through summaries, audio reading, quizzes, flashcards, study tips, and keywords.

The project addresses common student problems such as difficulty in reading handwritten notes, time-consuming revision, and lack of accessible study support. By using Python, Streamlit, PyTesseract, Tesseract OCR, Pillow, Ollama, and gTTS, the project provides a practical and useful prototype for AI-assisted learning.

Thus, the project meets its objective of creating a simple AI-powered platform that helps students learn more effectively from handwritten and image-based notes.

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